

YUVA INTERN

Virtual Data Science with Python Apprentice Intern

Week 4 Project Report

Machine Learning Model Development and Evaluation

TOPIC- *Heart Disease Prediction Using Machine Learning
Classification Models*

Dataset Used

cleaned_merged_heart_dataset.csv

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1. Introduction

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed. In this project, a machine learning classification model is developed to predict the presence of heart disease based on patient medical information.

The main objective of this project is to build, train, and evaluate machine learning models using Python. The complete workflow includes data preprocessing, exploratory data analysis, model development, performance evaluation, and error analysis.

The Heart Disease dataset is used to train classification models that can identify whether a patient is likely to have heart disease based on clinical parameters.

2. Dataset Description

The dataset used in this project is the **Heart Disease Dataset obtained from Kaggle**.

The dataset contains medical records of patients with different clinical attributes. The objective is to classify whether a patient has heart disease or not.

Dataset Details:

Parameter	Description
Dataset Type	Binary Classification
Number of Records	1888
Number of Features	13
Target Variable	target
Class 0	No Heart Disease
Class 1	Presence of Heart Disease

The dataset contains important medical features such as age, sex, chest pain type, cholesterol level, resting blood pressure, maximum heart rate, and other cardiovascular indicators.

3. Data Loading and Initial Analysis

The dataset was imported into Python using the Pandas library. Initial analysis was performed to understand the structure, dimensions, and statistical properties of the data.

Figure 1: Dataset Preview

	age	sex	cp	trestbps	chol	fbs	restecg	thalachh	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

Description:

The dataset contains patient information along with the target variable. The first few rows were examined to verify successful data loading and understand the available features.

Figure 2: Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1888 entries, 0 to 1887
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   age         1888 non-null   int64
1   sex         1888 non-null   int64
2   cp          1888 non-null   int64
3   trestbps    1888 non-null   int64
4   chol        1888 non-null   int64
5   fbs         1888 non-null   int64
6   restecg     1888 non-null   int64
7   thalachh    1888 non-null   int64
8   exang       1888 non-null   int64
9   oldpeak     1888 non-null   float64
10  slope       1888 non-null   int64
11  ca          1888 non-null   int64
12  thal        1888 non-null   int64
13  target      1888 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 206.6 KB
```

Description:

The dataset information shows the data types of different features and confirms the number of records and attributes used for model development.

4. Data Preprocessing

Before applying machine learning algorithms, preprocessing steps were performed to prepare the dataset.

The following steps were applied:

- Checked missing values
- Checked duplicate records
- Separated features and target variable
- Applied feature scaling using StandardScaler

Missing Value Analysis

Figure 3: Missing Values Check

(Insert screenshot of missing values output)

Description:

The dataset was checked for missing values. No missing values were detected, therefore no imputation method was required

```
[6]: age      0
     sex      0
     cp       0
     trestbps 0
     chol     0
     fbs      0
     restecg  0
     thalachh 0
     exang    0
     oldpeak  0
     slope    0
     ca       0
     thal     0
     target   0
     dtype: int64
```

The dataset does not contain missing values. Therefore, no missing value treatment is required.

5. Exploratory Data Analysis

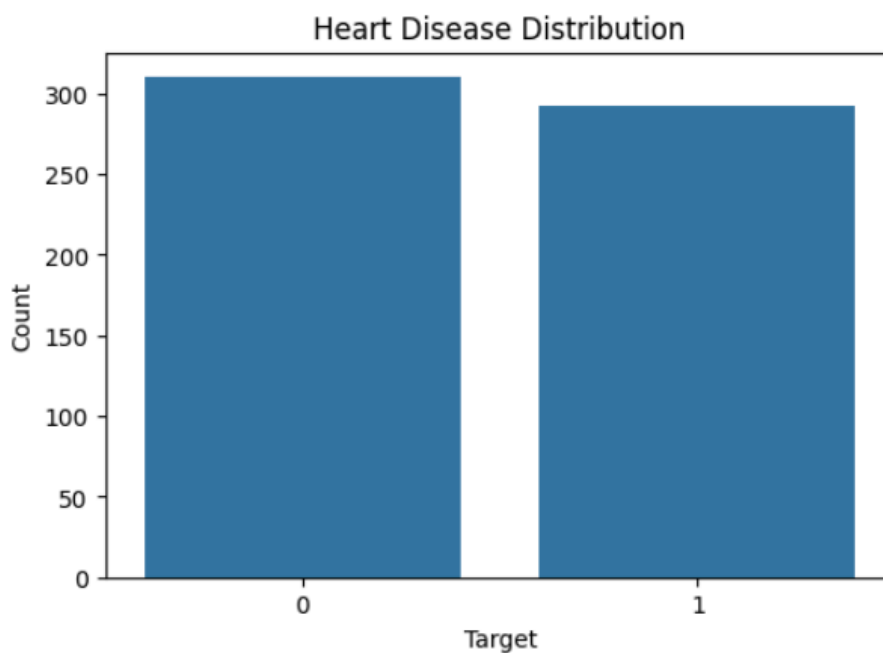
Exploratory Data Analysis was performed to understand patterns and relationships within the dataset.

5.1 Target Variable Distribution

Figure 4: Heart Disease Distribution

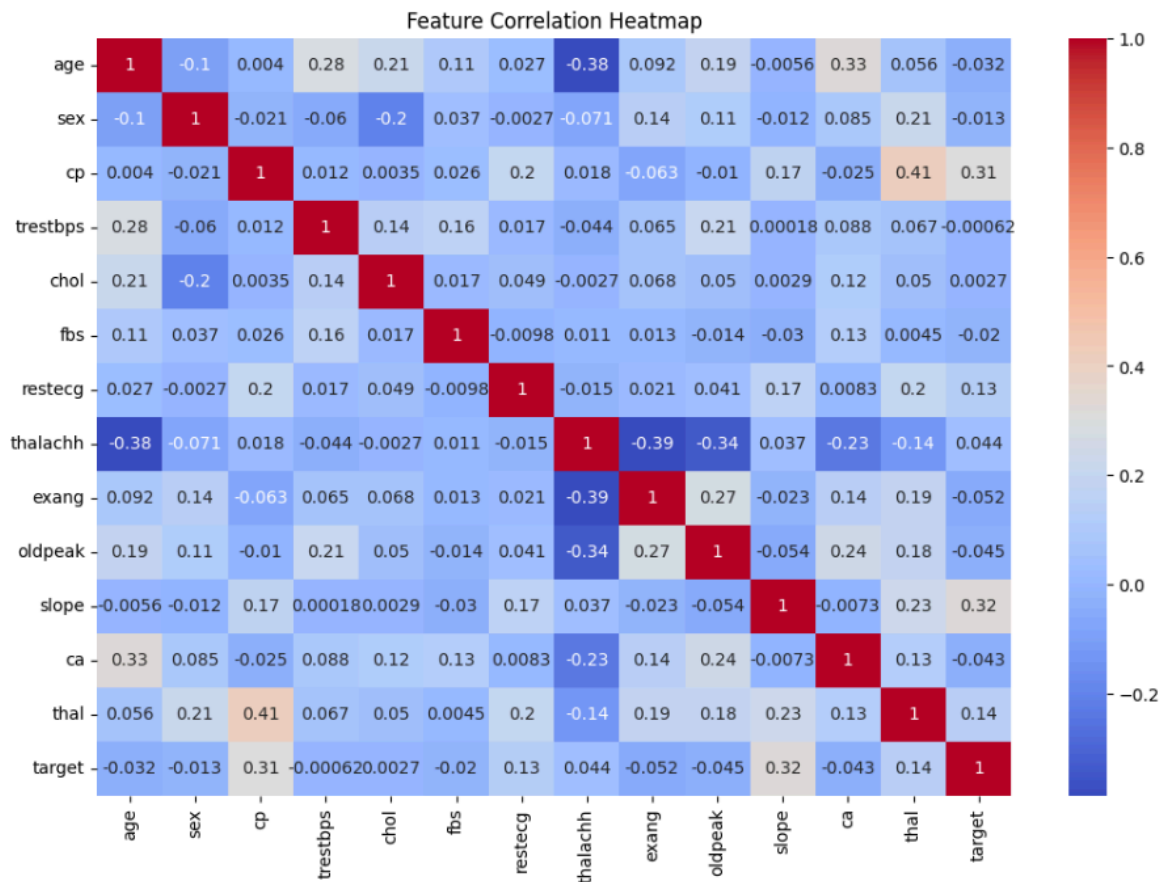
Description:

The target distribution visualization shows the number of patients belonging to each class. This helps understand the balance between patients with and without heart disease.



5.2 Feature Correlation Analysis

Figure 5: Correlation Heatmap



Description:

The correlation heatmap represents relationships between different medical features. It helps identify features that have stronger relationships with the target variable and may influence model predictions.

6. Feature Selection and Data Splitting

The dataset was divided into:

- Independent variables (features)
- Dependent variable (target)

The data was split into training and testing sets:

- 80% data used for training
- 20% data used for testing

Feature scaling was performed using StandardScaler to normalize numerical values and improve model performance

7. Machine Learning Model Development

7.1 Logistic Regression Model

Logistic Regression was implemented as the first classification algorithm.

Reason for Selection:

- Suitable for binary classification problems
- Simple and interpretable
- Provides probability-based predictions
- Efficient for medical prediction tasks

The model was trained using the training dataset and used to predict heart disease outcomes on unseen testing data.

7.2 Random Forest Classifier

Random Forest was implemented as a second model for comparison.

Reason for Selection:

- Handles complex relationships between features
 - Reduces overfitting
 - Provides feature importance information
 - Performs well on structured datasets
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8. Model Evaluation

The trained models were evaluated using:

Accuracy:

Measures the percentage of correct predictions.

Precision:

Measures the correctness of positive predictions.

Recall:

Measures how many actual heart disease cases were correctly detected.

F1-score:

Provides a balance between precision and recall.

9. Model Performance Comparison

Figure 6: Model Comparison Table

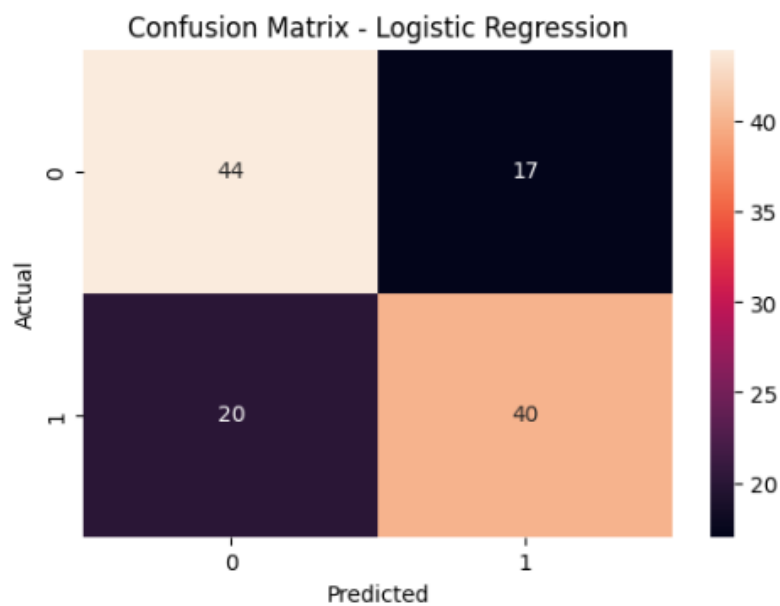
Description:

The comparison table shows the performance of Logistic Regression and Random Forest models based on different evaluation metrics.

The model with better performance can be selected as the final prediction model.

10. Confusion Matrix Analysis

Figure 7: Confusion Matrix



Description:

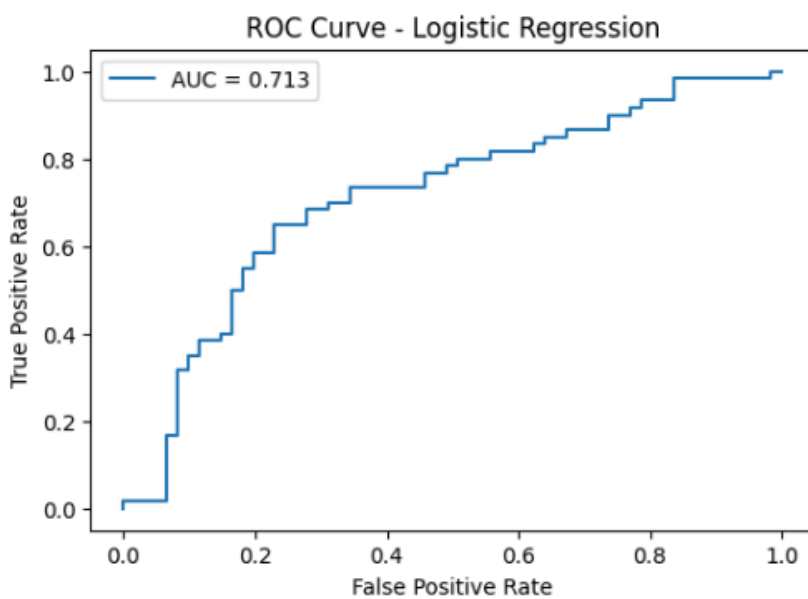
The confusion matrix evaluates the classification performance by showing:

- True Positive: Correctly predicted heart disease cases
- True Negative: Correctly predicted non-disease cases
- False Positive: Incorrect disease predictions
- False Negative: Missed disease cases

This analysis helps identify the types of errors made by the model.

11. ROC Curve Analysis

Figure 8: ROC Curve

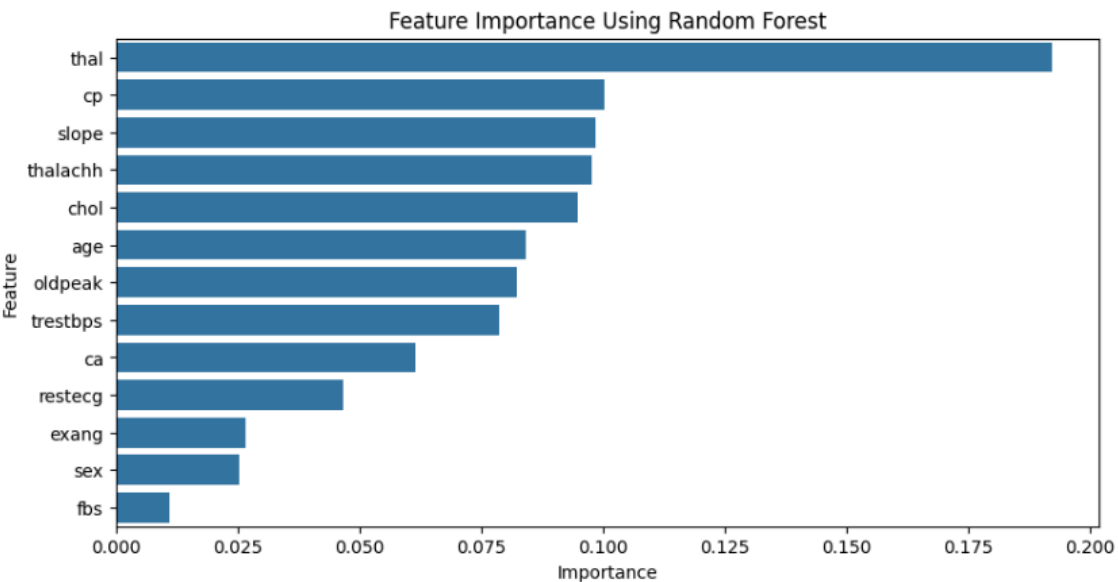
**Description:**

The ROC curve represents the ability of the model to distinguish between patients with and without heart disease.

A higher Area Under Curve (AUC) value indicates better classification performance.

12. Feature Importance Analysis

Figure 9: Feature Importance Graph



Description:

Feature importance analysis identifies which medical attributes contribute most to heart disease prediction.

Important features can help understand the factors influencing model decisions.

13. Error Analysis and Limitations

Although the developed models provide good prediction performance, some limitations exist:

Dataset Limitations:

- The dataset contains a limited number of patient records.
- It may not represent all demographic groups.

Feature Limitations:

- Lifestyle factors such as diet, exercise, and stress are not included.
- Additional medical parameters could improve prediction.

Model Limitations:

- Incorrect predictions may occur due to complex relationships between health factors.
 - The model requires validation on larger datasets before real-world usage.
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14. Future Improvements

The model can be improved by:

- Applying hyperparameter optimization
 - Using advanced algorithms such as XGBoost and Neural Networks
 - Performing cross-validation
 - Increasing dataset size
 - Applying feature selection techniques
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15. Conclusion

This project successfully implemented a machine learning-based heart disease prediction system.

The complete workflow included data preprocessing, exploratory analysis, model development, and evaluation. Logistic Regression and Random Forest models were trained and compared using multiple performance metrics.

The results demonstrate that machine learning techniques can effectively identify patterns in medical data and assist in predicting heart disease risk.